

and the same-member experiment isolated the uncertainty of XRO parameter fitting and its impact on reforecasting skill. All four LENS results using the same observational record length (43 year) confirmed that the uncertainty in parameter estimation leads to an XRO reforecasting correlation skill error of less than 0.1 within 21 lead months (Supplementary Fig. 8b–e).

Quantitative reforecasting experiments

To rigorously dissect the interplay between ENSO and the different climate modes in the different ocean basins, we designed three sets of sensitivity experiments to mimic the experiment protocol of previous CGCM studies:

1. Uninitialized experiments: we performed uninitialized mode- j experiments (U_j) by setting the initial condition of T_j to zero, while keeping everything else the same as in the control experiment. The effect of the mode- j initial condition can be assessed as the difference between the control and U_j (XRO- U_j). To disentangle the role of a specific ocean basin's initial conditions, we also conducted uninitialized experiments by setting the initial conditions of all modes to zero in the corresponding ocean basins. For example, the uninitialized ExPO experiment (referred to as U_{ExPO}) is the same as the control experiment but with the initial conditions of the NPMM and SPMM set to zero. Similarly, U_{IO} , U_{AO} and $U_{\text{ExPO+IO+AO}}$ denote the uninitialized IO, uninitialized AO and uninitialized 'all other basins' experiments, respectively. In addition, the uninitialized ENSO SSTA ($U_{\text{Nino3.4}}$) and WWV anomaly (U_{WWV}) experiments are same as XRO, except that the initial conditions of T_{ENSO} and h are set to zero, respectively. The uninitialized ENSO (U_{ENSO}) experiment is same as XRO, but the initial conditions of both T_{ENSO} and h are set to zero. The difference in the climate system response between the control experiment and U_j isolates the effect of mode- j and/or basin- j 's initialization.
2. Decoupled experiments: we performed decoupled mode- j experiments (referred to as D_j)—in which specific mode(s) are suppressed—by strongly increasing the diagonal damping rate of mode- j in the L operator to an e folding time scale of 5 days. This mimics the partially coupled experiments in fully coupled climate models that restore the ocean surface temperature towards prescribed conditions. The differences between the control experiment and D_j isolate the role of mode- j in the system. To disentangle the role of the different ocean basins, we conducted decoupled ocean basin experiments. For example, the decoupled ExPO experiment (referred to as D_{ExPO}) removes both the NPMM and SPMM from the system. Similarly, the decoupled IO experiment (D_{IO}) removes the IOB, IOD and SIOD together from the system; the decoupled AO experiment (D_{AO}) removes the TNA, ALT3 and SASD together from the system; and the decoupled all-other-modes experiment ($D_{\text{ExPO+IO+AO}}$) removes all other modes except ENSO. We note that the $D_{\text{ExPO+IO+AO}}$ experiment is very close to the nRO in which the parameters were fitted separately. The difference between the control experiment and D_j isolates the effect of mode- j and basin- j 's coupling. The sum of individual basin decoupled experiments exceeds the effect of decoupling all at once (Extended Data Fig. 3b,d,e), suggesting the presence of indirect pathways due to interactions among basins.
3. Relaxation towards observations experiments: we performed relaxation ocean basin- j experiments (referred to R_j) by relaxing the SSTA indices towards the observations in the corresponding ocean basins with a time scale of 5 days. For example, the relaxation ExPO experiment (referred to as R_{ExPO}) is the same as the control but with the NPMM and SPMM being relaxed to the observations. Similarly, R_{IO} , R_{AO} and $R_{\text{ExPO+IO+AO}}$ denote the relaxation IO, relaxation AO and relaxation all other basins except the equatorial Pacific experiments. The difference between the control experiment and R_j highlights the effect from perfect 'future' knowledge of basin- j . The relaxation

towards observations experiments greatly overestimates ENSO forecast skill because of built-in presumed perfect predictions for the stochastic excitations and ENSO's effects on the modes in these basins (magenta curves in Extended Data Fig. 3d,e).

Losing memory experiments

We carried out 'losing memory' experiments by artificially adding extra damping to the original diagonal damping rates of all other non-ENSO modes in the L_M operator (Extended Data Fig. 4). The prescribed damping rates are $(5 \text{ day})^{-1}$, $(30 \text{ day})^{-1}$, $(90 \text{ day})^{-1}$, $(180 \text{ day})^{-1}$ and $(360 \text{ day})^{-1}$, in the different experiments, ranging from strong damping (no memory) to less damping (long memory).

Deseasonalizing experiments

We carried out deseasonalizing experiments to illustrate the role of the operator parameters' annual and semi-annual cycles in ENSO forecast skill (Supplementary Fig. 17). In the XRO_{ac=0} model, we considered only the annual mean component ($\mathbf{L}_0$ and $\mathbf{N}_0$ in equations (4) and (5): each tendency equation had roughly ten parameters, so a total number of parameters of $103 = 10 \times 10 + 3$). Here, 10–15 years of data are required to meet the rule of thumb for regression sample size (at least ten subjects per predictor)⁷⁸. In the XRO_{ac=1} model, we considered both the annual mean and annual cycle components in the operator ($\mathbf{L}_0^c$, $\mathbf{L}_1^c$, $\mathbf{L}_1^s$, $\mathbf{N}_0^c$, $\mathbf{N}_1^c$ and $\mathbf{N}_1^s$ in equations (4) and (5), each tendency equation had roughly 30 parameters: the total number of parameters was $309 = 3 \times 100 + 3 \times 3$). At least 25 years of data are required⁷⁸. The difference between XRO and XRO_{ac=0} isolates the combined impacts of the annual and semi-annual cycles in the operator parameters, whereas the difference between XRO and XRO_{ac=1} isolates the impact of just the semi-annual cycle in the operator parameters. The parameters for the XRO_{ac=0} and XRO_{ac=1} experiments can be either refitted separately (Supplementary Fig. 17a–d) or taken from the XRO control experiment (Supplementary Fig. 17e–h). Regardless of which parameter estimation method is used, we found that the seasonal cycle is critically important in suppressing SPB for ENSO, whereas the semi-annual cycle is less important.

Removing nonlinearity experiments

We carried out 'removing nonlinearity' experiments to illustrate the role of the XRO nonlinear operators in ENSO forecast skill (Supplementary Fig. 18). In the XRO_{linear} experiment, we considered only linear operators and set $\mathbf{N}_{\text{ENSO}}$ and $\mathbf{N}_M$ to zero. In the XRO_{linearENSO} experiment, we only considered linear operators and $\mathbf{N}_M$, but set $\mathbf{N}_{\text{ENSO}}$ to zero. In the XRO_{linearIOD} experiment, we only considered linear operators and $\mathbf{N}_{\text{ENSO}}$, but set $\mathbf{N}_M$ to zero. The difference between XRO and XRO_{linear} isolates the impact of the nonlinear operator parameters, whereas the difference between XRO and XRO_{linearENSO} isolates the impact of the ENSO nonlinear operator parameters. The parameters for the XRO_{linear}, the XRO_{linearENSO} and XRO_{linearIOD} experiments can be either refitted separately (Supplementary Fig. 18a–d) or taken from the XRO control experiment (Supplementary Fig. 18e–h). Regardless of which method we used to obtain the parameters, we found that the ENSO nonlinear dynamics were critically important for ENSO forecast skill, especially for forecasting the amplitude of the peak phase and the fast transition from El Niño to La Niña. Further, we found that the impact of IOD's nonlinearity on ENSO forecast skill was neglectable.

Prediction skill metrics and significance tests

The forecast skill is quantified using the anomaly correlation coefficient (ACC) and root mean square error (r.m.s.e.) metrics⁹⁹. The ACC is computed as the Pearson correlation coefficient between the deterministic forecast (f) and the observations (o):

$$\text{ACC} = \frac{\text{cov}(f, o)}{\sigma_f \sigma_o}, \quad (7)$$